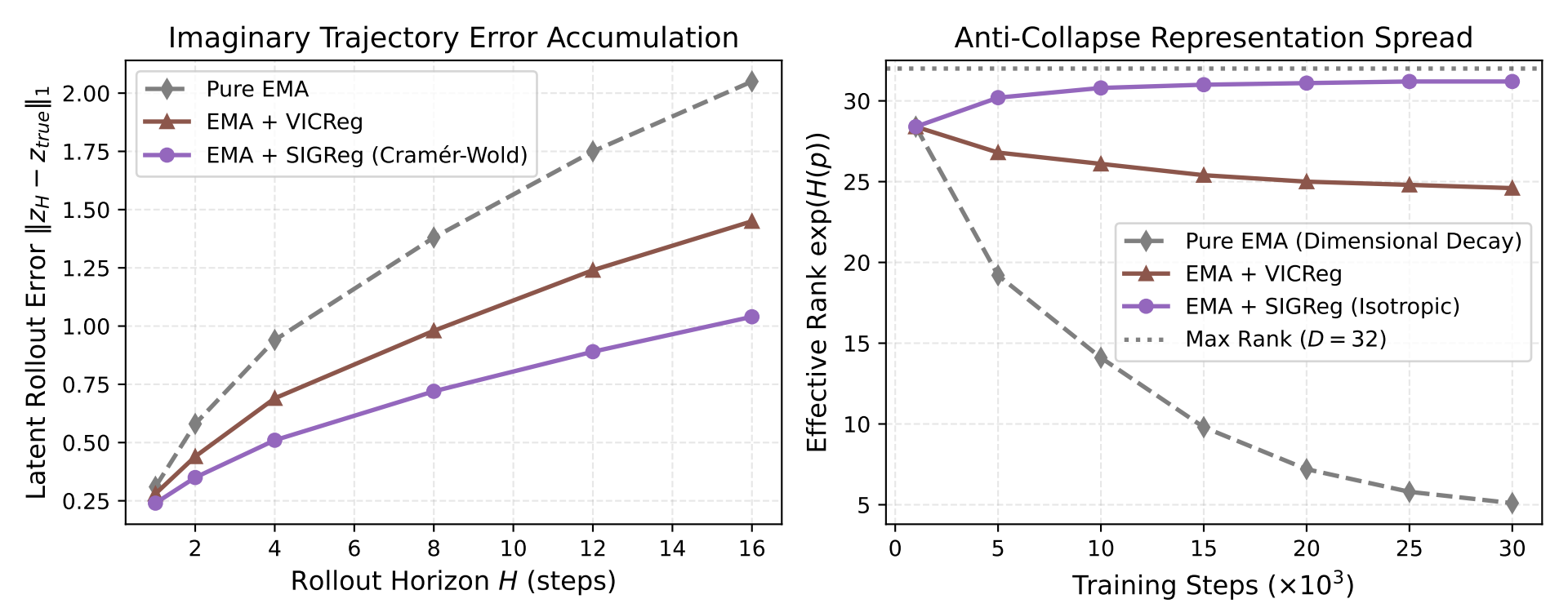
Imaginary Trajectory Error Accumulation
Latent Rollout Error $\|z_H - z_{true}\|_1$
Rollout Horizon H (steps)
Pure EMA
EMA + VICReg
EMA + SIGReg (Cramér-Wold)
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2
4
6
8
10
12
14
16
Anti-Collapse Representation Spread
Effective Rank $\exp(H(p))$
Training Steps ($\times 10^3$)
Pure EMA (Dimensional Decay)
EMA + VICReg
EMA + SIGReg (Isotropic)
Max Rank ($D = 32$)
5
10
15
20
25
30
0
5
10
15
20
25
30